

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
6	(a)	(i)		Black and [experiment] 2 (1) because it shows the higher temperature [so is emitting more IR] or <u>correct</u> numerical comparison i.e. 90 is bigger than 65 (1) Allow ecf on the experiment number provided black is chosen Award 0 marks if silver is chosen			2	2		2
		(ii)		Black and [experiment] 1 (1) because it shows the higher temperature [so is absorbing more IR] or <u>correct</u> numerical comparison i.e. 46 is bigger than 32 (1) Allow ecf on the experiment number provided black is chosen Award 0 marks if silver is chosen			2	2		2
	(b)			Silver surface is a poor emitter [of infra-red] / reflects the [radiated] <u>heat back in</u>		1		1		
	(c)	(i)		Payback time = $\frac{500}{25}$ (1) Payback time = 20 [years] (1)	1	1		2	2	
		(ii)		Loft insulation is cheaper than cavity wall insulation (1) but savings are the same (1) so disagree. Accept reverse argument OR The payback time for loft insulation is 3 years (1) compared to the 4 years for cavity wall insulation (1) so disagree OR Loft insulation is cheaper than cavity wall insulation (1) payback time for loft insulation is shorter (1) so disagree Conclusion needed to award 2 marks			2	2		
				Question 6 total	1	2	6	9	2	4